EpiScope

Bangladesh Disease Surveillance, Forecasting
and Resource Allocation

Technical Project Report

An academic prototype integrating district reference data, weather, synthetic health encounters, data quality, outbreak signal detection, short-term forecasting, resource allocation and an interactive D3.js dashboard.

Geographic scope: 64 districts of Bangladesh
Execution: Google Colab, Python and PySpark
Reference period: 2021–2025

Abstract

EpiScope is an academic prototype for district-level disease surveillance and decision support in Bangladesh. It combines district boundaries and population references, historical weather data, a synthetic facility registry and synthetic DHIS2-style encounter records. The workflow applies data-quality checks, quarantines invalid records, hashes patient identifiers before clean Silver storage and produces district-disease-week Gold summaries. A simple four-week baseline detector generated 64 alert records and achieved 22.58% event-level sensitivity against the synthetic outbreak plan. Seasonal Naive forecasts were evaluated for dengue and diarrhoea at one- to four-week horizons. A proportional allocation baseline was compared with an integer linear programming model using synthetic facility resource snapshots. A five-view D3.js dashboard presents aggregated outputs. The system is not a validated public-health surveillance service.

Keywords: disease surveillance, Bangladesh, PySpark, data quality, outbreak detection, forecasting, resource allocation, D3.js.

1. Introduction

Timely disease surveillance can help public-health teams identify unusual case patterns, understand geographic variation and plan limited resources. Such workflows require consistent geographic identifiers, reporting dates, quality controls and clear presentation. EpiScope demonstrates an academic workflow using Bangladesh district-level reference data and explicitly synthetic health records.

1.1 Objectives

The objectives are to integrate district, population and weather references; generate synthetic encounter and facility data; simulate reporting delays and backfill; validate and quarantine defective records; hash patient identifiers; create weekly summaries; demonstrate outbreak detection and forecasting; compare resource allocation approaches; and display aggregate results in a coordinated dashboard.

1.2 Scope and responsible use

Encounter, facility and stock datasets are synthetic. Outputs are for coursework and technical evaluation only. They must not be used for diagnosis, clinical decisions, real outbreak declarations or operational resource deployment.

2. System Architecture

The workflow follows layered data processing. Raw stores generated encounters and facility inputs. Reference stores district boundaries, population and weather features. Bronze contains quality reports and quarantined records. Silver stores clean encounters with hashed patient identifiers. Gold contains weekly summaries, outbreak alerts, forecasts and resource-allocation outputs. Aggregate files are served to the D3.js dashboard.

Component	Technology / format
Execution	Google Colab, Python, Java 17, PySpark
Data handling	Pandas, PyArrow, Parquet
Optimization	PuLP integer linear programming
Visualization	D3.js v7, HTML, aggregate CSV/GeoJSON
Weather	Open-Meteo historical archive

3. Data Sources and Preparation

District boundaries were obtained from a DGHS-hosted ArcGIS feature service. District population values came from an ArcGIS-hosted layer attributed to BBS 2022. Daily weather variables were retrieved from Open-Meteo for 64 district centroids over 2021–2025. Synthetic facilities and encounters were generated for the prototype.

Dataset	Verified coverage
District boundaries	64 district features; unique district identifiers
Population	64 districts; represented total 165,158,616
Weather	116,864 daily district records; 64 districts; 2021–2025
Facility registry	640 synthetic facilities; 10 per district

Weather data include daily mean temperature, precipitation and relative humidity. Derived features include prior rainfall windows, lagged temperature, extreme-rainfall thresholds and seasonal indicators. Disease-specific date weights shaped synthetic sampling. These coefficients are illustrative assumptions, not calibrated epidemiological parameters.

4. Synthetic Encounter Generation

The DHIS2-style prototype contains 10,200 rows, including deliberately introduced defects. The base prototype contained 10,000 records, with approximately 2% duplicate encounter identifiers, 0.5% orphan facility identifiers and 1% invalid ages. Disease-specific ICD-10 codes and outbreak labels were assigned from a separate ground-truth injection plan.

4.1 Outbreak and reporting simulation

The ground-truth plan contains 30 local events and one nationwide respiratory-surge event. The nationwide event is represented by 64 district-specific plan rows. The respiratory surge uses influenza-like illness and J10/J11 codes; it does not represent a real newly identified pathogen. Reporting delays range from 0 to 21 days. Weekend and selected holiday delays are simulated. Thirty-two facilities were selected for a 21-day silent period followed by backfill; six backfill records were observed.

The outbreak injection process is a prototype: it uses an assumed injection fraction and simple temporal curves, and assigns outbreak labels to existing baseline records rather than fully modelling a calibrated increase in incidence. It is therefore a controlled software test, not a realistic epidemiological benchmark.

4.2 Large-scale generation and benchmarks

Scale factor	Chunks	Rows incl. duplicates	Read (s)	Aggregation (s)
0.25	63	6,375,000	11.42	13.47
0.50	125	12,750,000	13.22	6.46
1.00	250	25,500,000	13.78	12.76

At SF=1, 25,000,000 base rows and 25,500,000 total rows were generated after adding 2% duplicates. All 250 Parquet chunks were verified. Spark read and aggregation benchmarks were run for all three scale factors. Timings are specific to the Colab environment; they compare read and aggregation only and do not establish optimization gains from caching or partition tuning.

The large-scale generator output does not include reporting_artifact and reporting_delay_days fields used by the 10,200-row prototype. Reporting-delay simulation was verified on the prototype, not on all 25 million base rows.

5. Data Quality and Privacy

Seventeen data-quality rules were applied to the 10,200-row prototype. Checks cover missing identifiers, unknown facilities, district consistency, dates, age, sex, disease-code compatibility, reporting delays, duplicate encounter identifiers and reporting-artifact values.

Quality issue	Observed records
Duplicate encounter IDs	200
Orphan facility IDs	50
Invalid ages	100
Quarantined unique rows	349
Clean Silver rows	9,851

The sum of rule failures is 350, while 349 unique rows were quarantined because one row failed two checks. Clean Silver records contain patient_id_hash rather than raw patient_id. The identifier is SHA-256 hashed using a user-supplied salt, which must be stored securely outside the repository. Hashing is pseudonymization, not a guarantee of complete anonymity.

Bronze metadata was created and integrity-checked for quality_reports (17 rows, 2 columns) and quarantine_encounters (349 rows, 19 columns). The original source-ingestion timestamp was not recorded, so the manifest leaves that field null; metadata creation time is not represented as ingestion time.

6. Silver and Gold Processing

Clean Silver contains 9,851 rows and 19 columns, covering 64 districts and four disease categories: dengue, diarrhoea, influenza-like illness and malaria. The Gold weekly summary contains 8,041 district-disease-week rows. A complete calendar for 64 districts, four diseases and 262 weeks gives 67,072 weekly records for detection.

Incremental processing was demonstrated with a 21-day lookback and repeat-run deduplication. A simulated 14-day reporting outage was recovered using the lookback window. Both tests preserved 9,851 unique clean records. These are Spark deduplication simulations, not a production Delta Lake MERGE implementation.

7. Outbreak Detection and Evaluation

The detector calculates a baseline from the previous four weeks and compares the current weekly count with the baseline mean plus two standard deviations. A minimum threshold of three cases is also applied. The corrected detector generated 64 alert records. This is a simple prototype threshold method, not a full implementation of EARS, CUSUM or Farrington algorithms.

Evaluation measure	Observed result
Ground-truth events	31
Detected events	7
Missed events	24
Event-level sensitivity	22.58%
False alert weeks	39
Median detection delay	21 days
False alert weeks per district-year	0.1219

Sensitivity is calculated at event level by matching alert weeks to the same district-disease event window. The false-alert rate is 39 unmatched alert weeks divided by 320 district-years (64 districts × 5 years). Low sensitivity indicates that the current detector misses many injected events. Threshold calibration and comparison with established methods are necessary before operational use.

8. Forecasting and Backtesting

Seasonal Naive forecasts were generated for dengue and diarrhoea at horizons of one to four weeks. The rolling backtest produced 3,328 predictions for each disease-horizon pair. Metrics include mean absolute error (MAE), nominal 80% prediction-interval coverage and weighted interval score (WIS).

Disease	Horizon	MAE	80% PI coverage	WIS
Dengue	1 week	0.3131	95.19%	0.2208
Dengue	2 weeks	0.3122	95.16%	0.2208
Dengue	3 weeks	0.3107	95.13%	0.2203
Dengue	4 weeks	0.3092	95.13%	0.2198
Diarrhoea	1 week	0.2266	95.79%	0.1696
Diarrhoea	2 weeks	0.2248	95.76%	0.1690
Diarrhoea	3 weeks	0.2242	95.85%	0.1683
Diarrhoea	4 weeks	0.2230	95.88%	0.1679

Coverage is substantially higher than the nominal 80%, suggesting conservative intervals. The WIS implementation is simplified and should be independently validated. Relatively low MAE is also influenced by sparse weekly counts and many zero values. No machine-learning forecasting comparison has been completed. The next-four-week forecast contains 512 rows (64 districts × 2 diseases × 4 horizons); no forecast is currently provided for influenza-like illness or malaria.

9. Resource Demand and Allocation

Demand was derived from forecast upper bounds and joined with a synthetic registry of 640 facilities. Synthetic snapshots contain available beds and disease-related test kits or supplies. These values are artificial and must not be interpreted as real facility inventory.

A proportional baseline distributed 494 demand units across 1,280 facility-disease allocation rows. An integer linear programming model minimized unmet demand with a large penalty and then minimized deviation from the proportional target. The solver returned an optimal solution with 494 units allocated and zero unmet demand. The LP changed 30 facility-disease allocation entries compared with the baseline, but did not increase total units served because the baseline also served all 494 units. Demand units are not clinical doses or medical recommendations.

10. Interactive Dashboard

The D3.js v7 dashboard contains five coordinated views: a district disease map, weekly disease trend, outbreak-alert view, four-week forecast view and resource-allocation comparison. District selection supports map zoom and highlighting. The browser-tested dashboard data package is approximately 1.08 MB, below the 10 MB aggregate-data target. It uses aggregate outputs and does not include patient-level identifiers.

11. Testing and Reproducibility

Seven automated pytest integrity tests passed with zero failures. They cover configuration presence and parsing, Silver privacy and uniqueness, facility registry integrity, weekly summary keys, forecast metrics and dashboard data constraints. A local Git repository and annotated v1.0 tag were created; publication to a remote Git hosting service has not been confirmed.

The project uses random seed 42 for synthetic generation. Reproducibility also requires securely retaining the patient hashing salt outside version control. A complete one-command end-to-end pipeline is not yet available, and production Delta MERGE remains unimplemented.

12. Limitations and Future Work

Area	Limitation / recommended work
Detection	22.58% event sensitivity; implement and compare EARS, CUSUM and Farrington; calibrate thresholds.
Forecasting	Seasonal Naive only for two diseases; compare models and validate intervals/WIS.
Synthetic epidemiology	Injection fractions and weather coefficients are illustrative; calibrate with suitable evidence.
Large-scale schema	25m base rows verified, but reporting fields were only simulated on the 10.2k prototype.
Incremental pipeline	Lookback and outage recovery are simulations; implement production Delta MERGE.
Bronze provenance	Manifest covers quality reports and quarantine; original ingestion timestamp was not captured.
Resource allocation	Facility and stock snapshots are synthetic; validate constraints with authorized operational data.
Packaging	Create reusable source modules and a reproducible end-to-end pipeline.

13. Conclusion

EpiScope demonstrates an integrated academic workflow for Bangladesh district-level surveillance, from reference-data preparation and synthetic encounter generation through quality control, privacy hashing, weekly aggregation, prototype detection, forecasting, resource allocation and dashboard presentation. The project verified SF=1 generation of 25 million base rows, three scale-factor Spark benchmarks, seven automated tests and a browser-tested dashboard. However, detector sensitivity, forecasting validation, synthetic resource inputs and simulated incremental processing show that the system remains a prototype. Further methodological validation, production-grade data engineering and responsible governance are required before real-world use.

References and Data Provenance

1. DGHS Bangladesh, district boundary reference layer hosted through the DGHS ArcGIS feature service: https://gis.dghs.gov.bd/server/rest/services/Hosted/bgd_admbnda_adm2_bbs_20201113/FeatureServer/0/query

2. ArcGIS Bangladesh hosted district population layer attributed to BBS 2022: https://www.arcgisbd.com/server/rest/services/ADB005/CRIIPS/MapServer/355/query

3. Open-Meteo, Historical Weather API: https://open-meteo.com/en/docs/historical-weather-api

4. D3.js documentation: https://d3js.org/

5. PuLP documentation: https://coin-or.github.io/pulp/

The synthetic encounter, outbreak, facility and stock datasets were generated for this project; they are not external clinical datasets.